



Major Project

HUMAN EMOTION DETECTION SYSTEM

Presented by:

Prince Yadav- R2142221078

Rajvardhan Singh- R2142220984

Mentored By- Dr Kamakshi Rautela



#RANKED 52
IN INDIA

#University Category



NO.1 PVT. UNIVERSITY IN
ACADEMIC REPUTATION IN INDIA



ACCREDITED **GRADE 'A'**
BY NAAC



PERFECT SCORE OF **150/150** AS A TESTAMENT
TO EXCEPTIONAL E-LEARNING METHODS



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INTRODUCTION

- Human emotions play a vital role in communication, decision-making, and behavior.
- Understanding emotions from **facial expressions, speech, and language** is essential for building intelligent human-centric systems.
- Traditional emotion detection approaches rely on **single data modalities**, which often leads to inaccurate and incomplete analysis.
- With advancements in **Deep Learning, Natural Language Processing, and Large Language Models**, it is now possible to analyze emotions more accurately using **multimodal data**.
- This project proposes **EmoSense**, a **Multimodal Emotion Detection System** that analyzes **video, audio, and speech text** to predict human emotional states and aggression levels, while also providing **human-readable explanations**.

MOTIVATION

- Growing demand for **human-aware intelligent systems**.
- Increasing applications in **mental health, security, interviews, and human–computer interaction**.
- Advancements in **Deep Learning and Large Language Models** enable better emotion understanding.
- Motivation to build a system that is:
 - More accurate than traditional methods
 - Multimodal (video + audio + text)
 - Explainable using LLMs
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LITERATURE REVIEW

Related work-

1. Multimodal Emotion Recognition (MER) surveys show that shallow fusion methods (early/late fusion) fail to handle modality imbalance and noise, motivating hierarchical architectures.
2. Video-based emotion recognition reviews highlight that uniform temporal pooling suppresses micro-expressions, recommending adaptive temporal modeling.
3. Audio–visual emotion surveys confirm that speech and facial expressions are temporally asynchronous, requiring explicit alignment mechanisms.
4. Multimodal fusion review papers demonstrate that deep cross-attention outperforms concatenation, but static fusion weights cause modality dominance.
5. Text-based emotion analysis surveys emphasize that text provides semantic context and should be fused at higher levels, not during perceptual feature extraction.
6. Recent surveys on LLMs in affective computing conclude that LLMs are best suited for high-level semantic reasoning over fused representations, not raw multimodal perception.

PROBLEM STATEMENT

- Human emotions are expressed through **face, voice, and spoken words**.
- Most existing emotion detection systems use **only one type of data** (text, audio, or video).
- Single-modal systems are **less accurate** and fail to detect **complex emotions like aggression**.
- Many systems do not provide **clear explanations** for their predictions.
- Therefore, there is a need for a **multimodal, accurate, and explainable emotion detection system**.

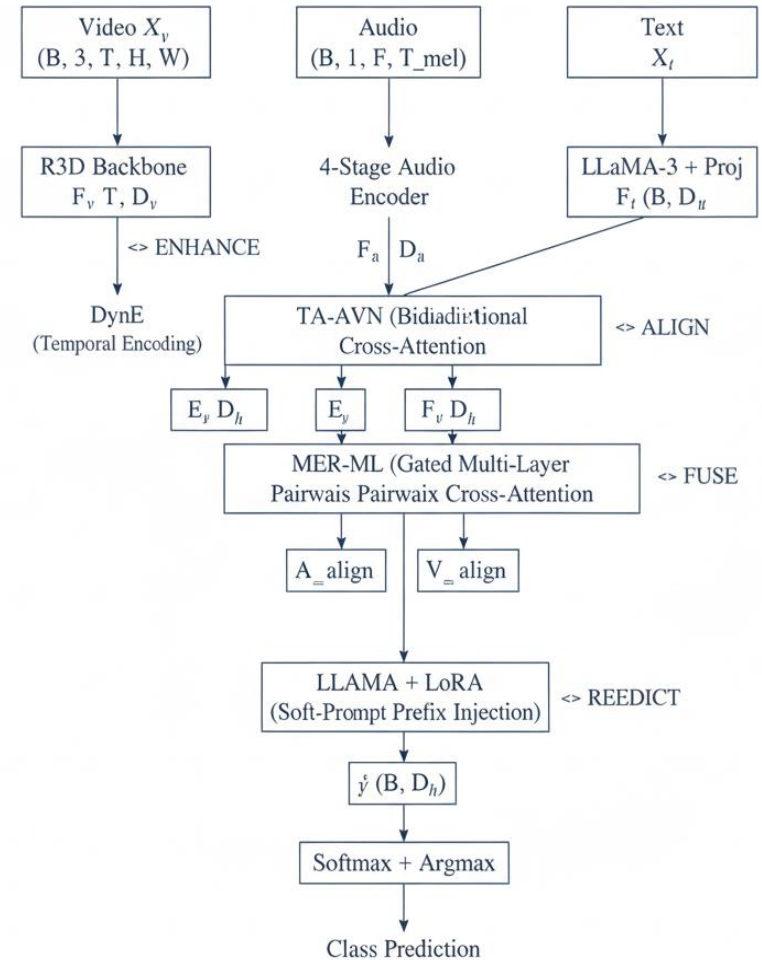
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OBJECTIVE

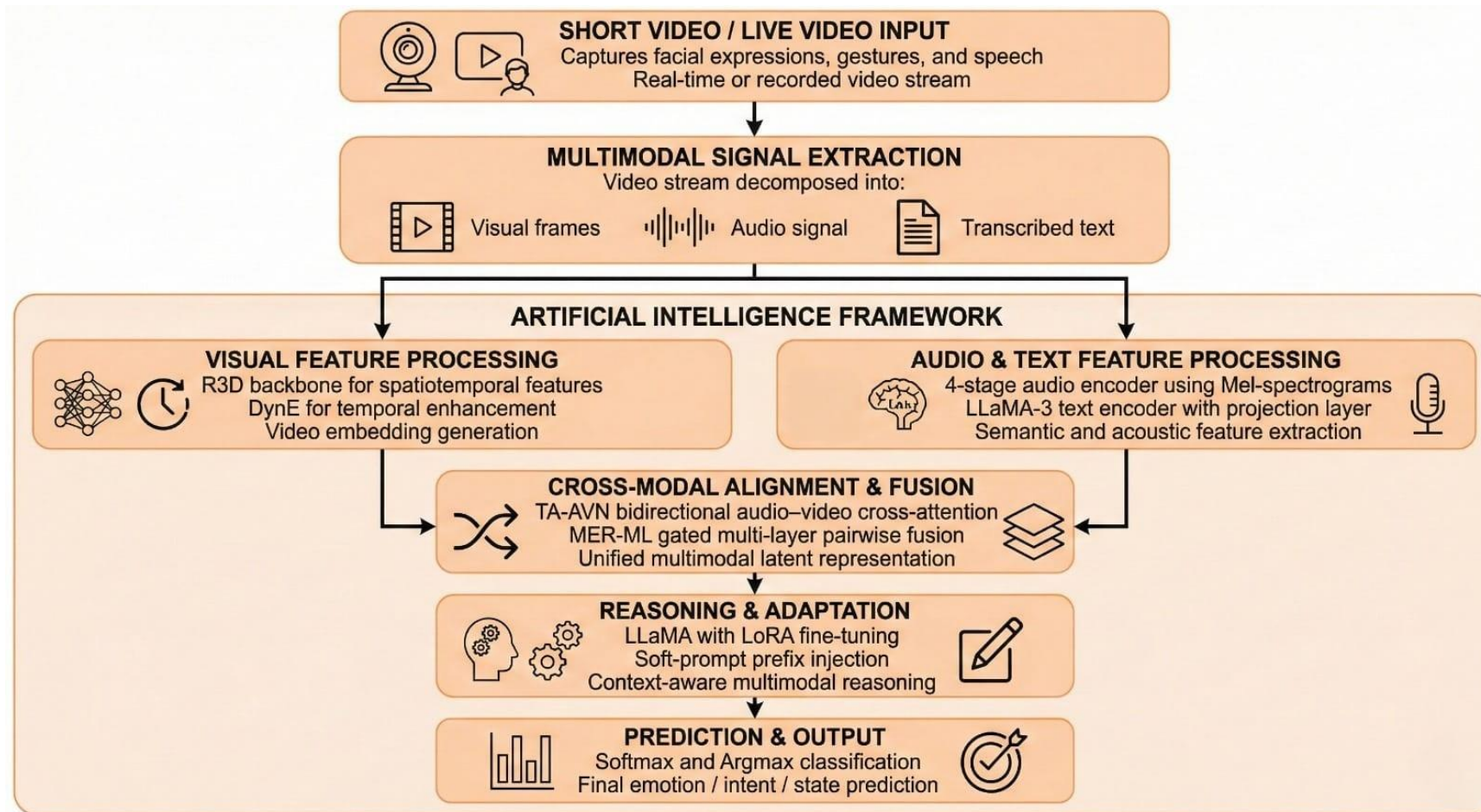
- To design a **multimodal emotion detection system** using video, audio, and speech text.
- To detect **human emotional states** such as happy, sad, angry, and neutral.
- To identify **aggression levels** using speech and facial cues.
- To combine outputs from multiple modalities to **improve accuracy**.
- To generate **human-readable emotion explanations** using a Large Language Model.

METHODOLOGY

- Collect emotion-based **video and audio data**.
- Extract **facial features** from video frames for emotion detection.
- Extract **speech features** from audio to analyze emotion and aggression.
- Convert speech to text and perform **NLP-based emotion analysis**.
- Combine video, audio, and text results using **multimodal fusion**.
- Generate **final emotion prediction and explanation** using an LLM.

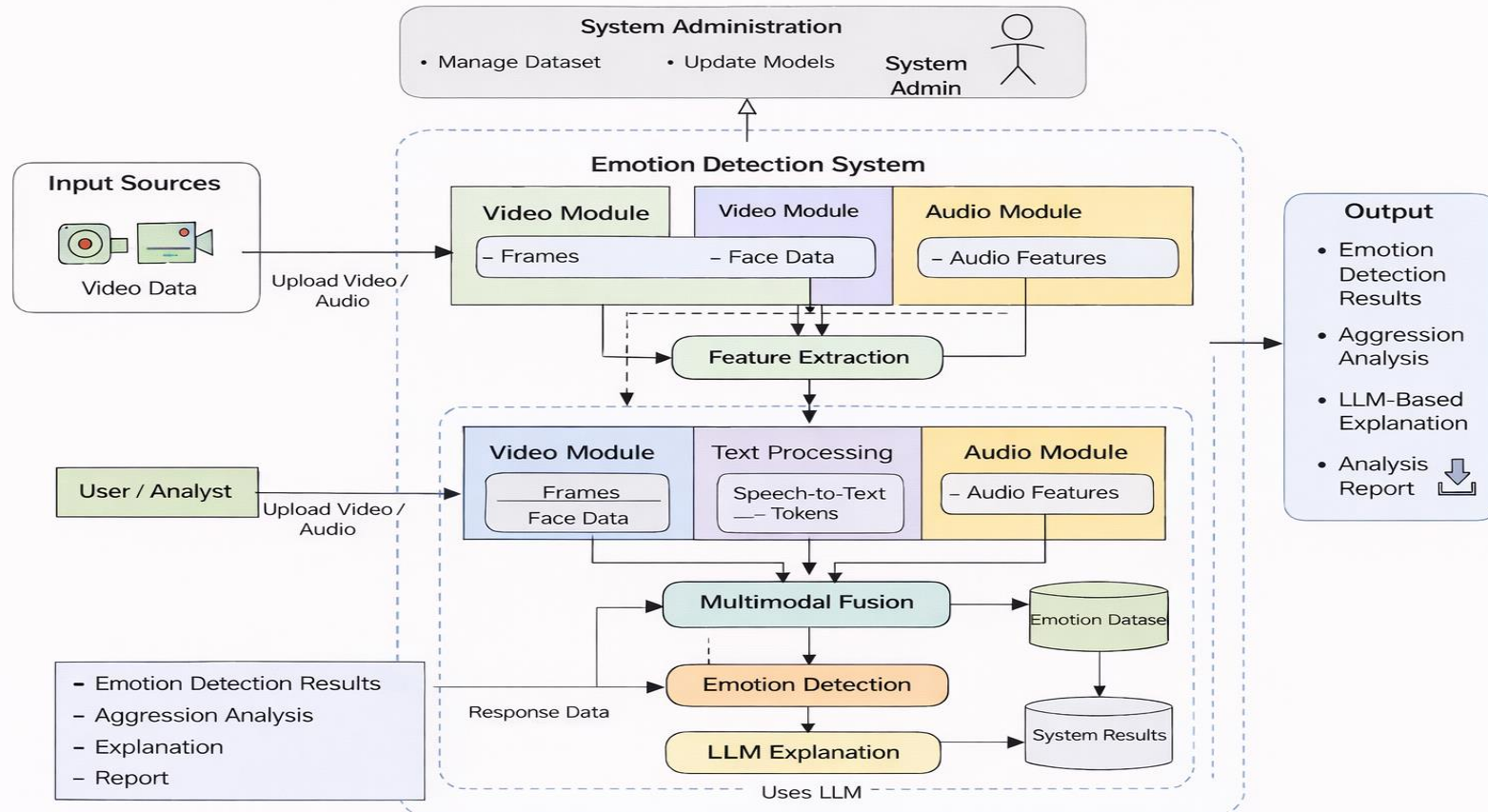


METHODOLOGY



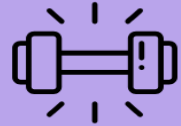
WORKING MODULE

System Architecture - Multimodal Emotion Detection (Sentivue)



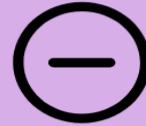
SWOT ANALYSIS

TECHNOLOGY SOLUTIONS



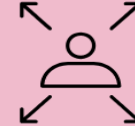
STRENGTHS

- Uses **multimodal data** (video, audio, and text) for higher accuracy.
- Integrates **Deep Learning and LLMs** for intelligent and explainable outputs.
- Provides **emotion and aggression detection**, not just sentiment.
- Modular architecture allows easy scalability and upgrades.
- Applicable across multiple real-world domains.



WEAKNESSES

- Requires **high-quality audio and video data** for best performance.
 - Computationally intensive due to deep learning models.
 - Performance may degrade in **noisy or low-light environments**.
- Large datasets are needed for effective training..



OPPORTUNITIES

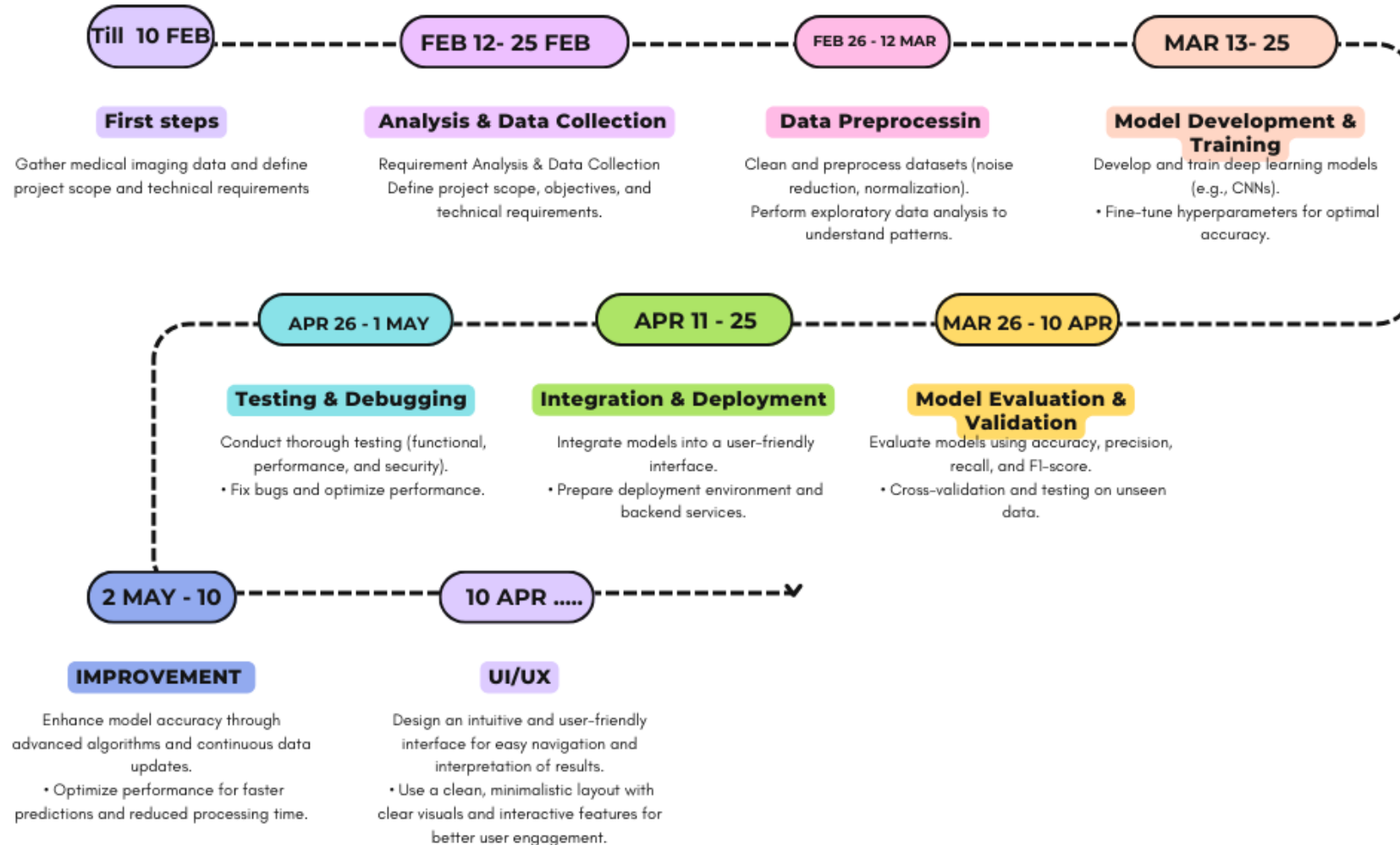
- Growing demand for **emotion-aware AI systems**.
- Strong potential in **mental health, HR, security, and healthcare**.
- Can be extended to **real-time emotion detection**.
- Scope for integration with **IoT, wearables, and smart devices**.
- High potential for **research publications and startup applications**.



THREATS

- **Privacy and ethical concerns** related to emotion monitoring.
- Variations in emotion expression across cultures.
- Legal and regulatory challenges in surveillance-based applications.
- Competition from big tech emotion analytics platforms.

Timeline



AREA OF APPLICATION

1. Mental Health Monitoring

Helps in identifying stress, depression, and emotional instability through speech and facial analysis.

2. Security and Surveillance Systems

Detects aggressive or suspicious behavior in public places using audio-visual emotion cues.

3. Interview and HR Assessment

Analyzes candidate emotions, confidence, and stress levels during interviews.

4. Customer Service and Call Centers

Monitors customer emotions and aggression to improve service quality and response strategies.

5. Human–Computer Interaction

Enables emotion-aware virtual assistants and intelligent systems.

6. Online Education and E-Learning

Tracks student engagement, attention, and emotional response during lectures.

7. Social Media and Content Moderation

Identifies aggressive or emotionally sensitive content.

8. Healthcare and Telemedicine

Assists doctors in understanding patient emotional states during remote consultations.

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THANK YOU

